



Leaf-inspired electronics innovation

A research team at TU Dresden has pioneered 'Leaftronics,' a sustainable electronics innovation inspired by natural leaf structures. Addressing over 60 million tons of annual



Lignocellulose-reinforced polymer films in leaves withstanding the manufacturing process for soldered circuitry (Credit: Kai Schmidt)

e-waste, Leaftronics introduces biodegradable polymer substrates as an alternative to non-recyclable printed circuit boards made from epoxy resin. Utilising quasi-fractal lignocellulose structures from leaves, these substrates achieve high thermal stability and biodegradability,

overcoming previous challenges. Leaftronics offers three times the sustainability of paper-based substrates, enabling eco-friendly disposal and material recovery. This holds transformative potential for consumer electronics, renewable energy, and research industries, aligning with global green technology goals.

Revolutionary shape-shifting, 3D printed antennas

Johns Hopkins University's Applied Physics Laboratory (APL) researchers have developed revolutionary shape-shifting antennas using 3D printing and shape memory alloys like nitinol. These antennas dynamically alter their form with temperature changes, enabling operation across multiple radio-frequency bands. Unlike traditional fixed-shape antennas, APL's design offers enhanced flexibility for military, scientific, and commercial applications, such as spectrum management and long- to short-range communication. The team overcame challenges in heating and maintaining RF properties by engineering a novel power line. Capable of transitioning between flat and cone shapes, this innovation is being patented and heralded as transformative for compact, adaptive communication systems.

X-shaped, 3D printed robot for diverse environments

A team of South Korean roboticists and engineers, in collaboration with the Max Planck Institute for Intelligent Systems, has developed a soft, X-shaped robot for handling both delicate and rigid objects in diverse environments. Constructed from a polydimethylsiloxane (PDMS) matrix embedded with NdFeB magnetic particles, the robot was fabricated using 3D printing and casting techniques. Magnetic control enables

precise movements, while octopus-inspired suction grippers enhance adhesion in wet, oily, or gooey conditions. Inspired by the velvet worm, it uses magnetorheological elastomers to adjust adhesive properties. Applications include surgical tasks, food processing, and electronics, showcasing a significant leap in bioinspired robotics.

New device to replace bulky infrared detectors

Researchers at CIC nanoGUNE in Spain have developed an on-chip device that identifies molecules by analysing their vibrational 'fingerprints' using hyperbolic phonon polaritons in a graphene-based infrared detector. This enhances sensitivity, enabling the detection of nanometer-thin molecular

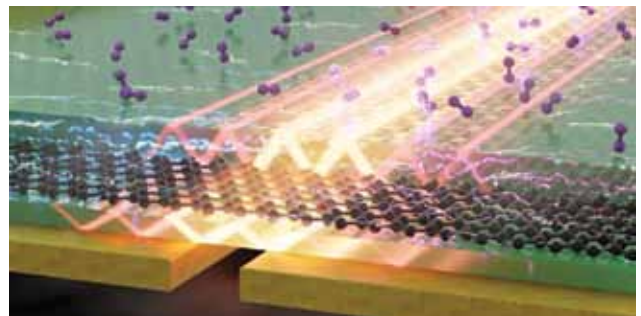


Illustration of an on-chip molecular vibration sensor based on a graphene IR detector, where phonon polaritons (bright rays) enhance the molecular fingerprint signal encoded in the photocurrent (Credit: Dr David Alcaraz, ICFO)

layers at room temperature and overcoming challenges faced by traditional infrared spectroscopy. Advances in surface-enhanced infrared absorption (SEIRA) amplify signals using metallic nanostructures. The device integrates SEIRA with graphene detectors for electrical detection, eliminating bulky infrared detectors. Researchers collaborating with ICFO and theoretical physics centres achieved the first on-chip SEIRA detection of molecular vibrations. This compact technology promises transformative diagnostics and environmental monitoring applications, potentially integrating with smartphones and wearables for portable spectroscopy.

Ultra-sensitive and durable hydrogen sensor

The Tokyo's Institute of Science team has developed a hydrogen sensor capable of swiftly detecting as low as five ppb concentrations. Using nano-patterned CuO nanowires on a silicon substrate with titanium and platinum electrodes, the sensor detects changes in resistance from hydrogen interactions. Advanced fabrication techniques, like electron-beam lithography and ex-situ oxidation, enhance sensitivity and durability. A 33nm electrode gap boosts detection within seven seconds, maintaining precision even in humidity.